Evaluation Report

Rule-Based POS Tagger for Technical Repair Transcripts

Purpose. Evaluate the rule-based POS tagger on its original annotated set and a separate eight-sentence challenge set, then test whether changing regex rule priority changes predictions.

1. Evaluation setup

Token accuracy is the number of correctly tagged tokens divided by the number of gold-standard tokens. The original set contains 8 sentences and 60 tokens. The additional challenge set contains 8 sentences and 47 tokens. The added examples target ambiguous technical terms and suffix heuristics. These are manually annotated, constructed examples rather than a representative sample of real-world repair transcripts.

2. Results

Evaluation set	Sentences	Tokens	Correct	Token accuracy
Original annotated set	8	60	60/60	100.00%
Additional challenge set	8	47	43/47	91.49%
Combined, production order	16	107	103/107	96.26%

3. Errors found in the additional challenge set

Token / context	Gold tag	Predicted tag	Likely cause
cable Wire the replacement cable.	NOUN	ADJ	Suffix heuristic .*able$ assigns ADJ, but cable is a noun here.
connects The wire connects the pump.	VERB	NOUN	Verb inflection -s is not covered by the current morphology rules or lexical list.
reads The gauge reads zero.	VERB	NOUN	Verb inflection -s is not covered by the current morphology rules or lexical list.
required The required torque is 20 Nm.	ADJ	VERB	The broad .*ed$ rule assigns VERB, but required is adjectival in this context.

4. Rule-order sensitivity

Four priority configurations were compared on the combined 16-sentence / 107-token set. The production configuration uses lexical rules before morphology, then the default fallback, followed by contextual corrections.

Configuration	Correct / total	Accuracy
lexical_first	103/107	96.26%
morphology_first	102/107	95.33%
reverse_lexical	103/107	96.26%

Configuration	Correct / total	Accuracy
reverse_morphological	103/107	96.26%

The morphology-first configuration changed “damaged” from ADJ to VERB because the -ed suffix rule took priority over the exact lexical rule. Its combined accuracy dropped from 96.26% to 95.33%. Reversing the order within the lexical or morphological lists did not change predictions on this dataset. This shows that cross-group priority can matter even when the order within a group has no observed effect on the tested examples.

5. Conclusions and limitations

• The original 60-token demonstration set passes completely, but the separate challenge set exposes four errors (91.49% accuracy).

• Common failure modes are overgeneralized suffix rules, missing third-person singular verb forms, and words whose grammatical role depends on context.

• The fallback NOUN tag is useful for technical objects but can misclassify unknown verbs and adjectives.

• The results are evidence about these 16 constructed sentences only. A larger held-out collection of authentic transcripts should be annotated before making broad performance claims.

Reproducibility: run `python src/pipeline.py` for the original set and `python src/evaluation_analysis.py` for challenge-set evaluation and rule-order sensitivity.